

SWEN90010 - High Integrity Systems Engineering

Hoare Logic

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Previous Lecture



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Statements

$S \quad := \quad \text{skip} \mid v := E \mid a[NE] := E \mid p(E_1, \dots, E_n) \mid S_1; S_2 \mid$
 $\quad \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ endif} \mid \text{while } B \text{ do } C \text{ done}$

where v is a variable, a is an array variable, and p is a procedure name.

$$\{P\} S \{Q\}$$

Hoare Triple

Previous Lecture



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Hoare Triple

$$\{P[E/x]\} x := E \{P\}$$

weakest pre-condition axiom
of assignment statement for
backward reasoning

Previous Lecture



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$$\frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

consequence rule for
two Hoare triples

Previous Lecture



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$$\frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

consequence rule for
two Hoare triples

also, strong implies weak

Combining the Two Rules

$$\frac{}{\{P[E/x]\} \ x := E \ \{P\}}$$

Assignment axiom

$$\frac{P' \Rightarrow P, \ Q \Rightarrow Q', \ \{P\} \ S \ \{Q\}}{\{P'\} \ S \ \{Q'\}}$$

Consequence rule

General pattern to prove triples about assignment that are not in the form of assignment axiom:

$$\{x = 0\} \ x := x + 1 \ \{x > 0\}$$

Combining the Two Rules

$$\frac{}{\{P[E/x]\} \ x := E \ \{P\}} \qquad \frac{P' \Rightarrow P, \ Q \Rightarrow Q', \ \{P\} \ S \ \{Q\}}{\{P'\} \ S \ \{Q'\}}$$

Assignment axiom

Consequence rule

General pattern to prove triples about assignment that are not in the form of assignment axiom:

1. apply consequence rule first to generalise precondition
2. then apply other rules (for example, assignment axiom)

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Combining the Two Rules



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Assignment axiom

Consequence rule

General pattern to prove triples about assignment that are not in the form of assignment axiom:

1. apply consequence rule first to generalise precondition
2. then apply other rules (for example, assignment axiom)

$$\frac{x = 0 \Rightarrow x + 1 > 0 \quad \{x + 1 > 0\} \ x := x + 1 \ \{x > 0\}}{\{x = 0\} \ x := x + 1 \ \{x > 0\}}$$

Combining the Two Rules

$$\frac{}{\{P[E/x]\} x := E \{P\}} \qquad \frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

Assignment axiom

Consequence rule

1. apply consequence rule first to generalise precondition:

$$\frac{x = 0 \Rightarrow ?P \qquad \{ ?P \} x := x + 1 \{ x > 0 \}}{\{ x = 0 \} x := x + 1 \{ x > 0 \}}$$

As a result, we have two goals to prove

Combining the Two Rules

$$\frac{}{\{P[E/x]\} x := E \{P\}}$$

Assignment axiom

$$\frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

Consequence rule

Now apply the assignment axiom

$$\frac{x = 0 \Rightarrow ?P \quad \{ ?P \} x := x + 1 \{ x > 0 \}}{\{ x = 0 \} x := x + 1 \{ x > 0 \}}$$

Combining the Two Rules



$$\frac{}{\{P[E/x]\} x := E \{P\}}$$

Assignment axiom

$$\frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

Consequence rule

$$x = 0 \Rightarrow x + 1 > 0 \quad \frac{}{\{x + 1 > 0\} x := x + 1 \{x > 0\}}$$

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Combining the Two Rules

$$\frac{}{\{P[E/x]\} x := E \{P\}}$$

Assignment axiom

$$\frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

Consequence rule

$$\frac{}{x = 0 \Rightarrow x + 1 > 0}$$

$$\frac{}{\{x + 1 > 0\} x := x + 1 \{x > 0\}}$$

$$\{x = 0\} x := x + 1 \{x > 0\}$$

Assignment Rule as Weakest Precondition Rule



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Given some **concrete postcondition P**,
the assignment rule says how to **calculate**
the **weakest precondition** to guarantee P holds afterwards:

$$\{ \quad \} x := E \{ P \}$$

Assignment Rule as Weakest Precondition Rule



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Assignment Rule as Weakest Precondition Rule

Given some **concrete postcondition P**,
the assignment rule says how to **calculate**
the **weakest precondition** to guarantee P holds afterwards:

$$\{P[E/x]\} x := E \{P\}$$

The rules we will see for the other kinds of statements
(if, while, skip, sequencing etc) will also be of this form,
allowing us to **calculate weakest preconditions** from
a **concrete postcondition**

Weakest Precondition Reasoning

Given concrete conditions P' and Q and some program S ,
to prove:

$$\{P'\} S \{Q\}$$

Weakest Precondition Reasoning

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First apply the consequence rule to generalise the
precondition, giving us two goals to prove:

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Given concrete conditions P' and Q and some program S ,
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First apply the consequence rule to generalise the
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$$\{?P\} S \{Q\}$$

$$P' \implies ?P$$

Weakest Precondition Reasoning

Given concrete conditions P' and Q and some program S ,
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First apply the consequence rule to generalise the
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Then, calculate what $?P$
is by applying the
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rules

Weakest Precondition Reasoning

Given concrete conditions P' and Q and some program S ,
to prove:

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First apply the consequence rule to generalise the
precondition, giving us two goals to prove:

$$\{?P\} S \{Q\}$$

$$P' \implies ?P$$

Then, calculate what $?P$
is by applying the
weakest precondition
rules

Once $?P$ has been
calculated, finally
check this
implication holds

Rule of Sequencing



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$$\frac{\{P\} S_1 \{R\}, \{R\} S_2 \{Q\}}{\{P\} S_1; S_2 \{Q\}}$$

Rule of Sequencing



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$$\frac{\{P\} S_1 \{R\}, \{R\} S_2 \{Q\}}{\{P\} S_1; S_2 \{Q\}}$$

Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

Rule of Sequencing



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Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

Proof:

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$$\frac{\{P\} S_1 \{R\}, \quad \{R\} S_2 \{Q\}}{\{P\} S_1; S_2 \{Q\}}$$

Apply consequence rule first

Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

Proof:

$$\{x = 0\} x := x+1; x := x+1 \{x=2\}$$

Rule of Sequencing



$$\frac{\{P\} S_1 \{R\}, \{R\} S_2 \{Q\}}{\{P\} S_1; S_2 \{Q\}}$$

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Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

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Rule of Sequencing



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Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

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Rule of Sequencing



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Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

Proof:

$$\frac{x = 0 \Rightarrow ?P \quad \frac{\{?P\} x := x+1 \{?R\} \quad \{?R\} x := x+1 \{x=2\}}{\{?P\} x := x+1; x := x+1 \{x=2\}}}{\{x = 0\} x := x+1; x := x+1 \{x=2\}}$$

Rule of Sequencing



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Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

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$$\frac{\{P\} S_1 \{R\}, \{R\} S_2 \{Q\}}{\{P\} S_1; S_2 \{Q\}}$$

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Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

Proof:

$$\{?P\} x := x+1 \{x+1=2\}$$

$$\{x+1=2\} x := x+1 \{x=2\}$$

$$x = 0 \Rightarrow ?P$$

$$\{?P\} x := x+1; x := x+1 \{x=2\}$$

$$\{x = 0\} x := x+1; x := x+1 \{x=2\}$$

Rule of Sequencing



$$\frac{\{P\} S_1 \{R\}, \{R\} S_2 \{Q\}}{\{P\} S_1; S_2 \{Q\}}$$

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$$\frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

Proof:

$$\frac{}{\{x+2=2\} x := x+1 \{x+1=2\}} \quad \frac{}{\{x+1=2\} x := x+1 \{x=2\}}$$

$$\frac{x=0 \Rightarrow x+2=2 \quad \{x+2=2\} x := x+1; x := x+1 \{x=2\}}{\{x = 0\} x := x+1; x := x+1 \{x=2\}}$$

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Theorem: $\{x = 0\} x := x+1; x := x+1 \{x=2\}$

Proof:

$$\frac{\frac{}{\{x+2=2\} x := x+1 \{x+1=2\}} \quad \frac{}{\{x+1=2\} x := x+1 \{x=2\}}}{\frac{}{\{x+2=2\} x := x+1; x := x+1 \{x=2\}}} \quad \frac{}{x=0 \Rightarrow x+2=2}$$

$$\{x = 0\} x := x+1; x := x+1 \{x=2\}$$

Larger Programs



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$$\begin{array}{l} \{x = X \wedge y = Y\} \\ t := x; \\ x := y; \\ y := t \\ \{ \end{array}$$

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$$\{x = X \wedge y = Y\}$$
$$t := x;$$
$$x := y;$$
$$y := t$$
$$\{x = Y \wedge y = X\}$$

Larger Programs

$$\{x = X \wedge y = Y\}$$
$$t := x;$$
$$x := y;$$
$$y := t$$
$$\{x = Y \wedge y = X\}$$

How to use sequence rule to reason over this?

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$$\begin{array}{l} \{?P\} \\ t := x; \\ x := y; \\ y := t \\ \{x = Y \wedge y = X\} \end{array}$$
$$x = X \wedge y = Y \Rightarrow ?P$$

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$$x = X \wedge y = Y \Rightarrow ?P$$
$$\begin{array}{l} \{?P\} \\ t := x; \\ \left(\begin{array}{l} x := y; \\ y := t \end{array} \right) \\ \{x = Y \wedge y = X\} \end{array}$$

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$$x = X \wedge y = Y \Rightarrow ?P$$
$$\begin{array}{l} \{?P\} \\ t := x; \\ \left(\begin{array}{l} x := y; \\ y := t \end{array} \right) \\ \{x = Y \wedge y = X\} \end{array}$$

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$$\frac{\begin{array}{l} \{?P\} \\ t := x; \\ \{?R_1\} \end{array}}{\begin{array}{l} \{?R_1\} \\ x := y; \\ y := t \\ \{x = Y \wedge y = X\} \end{array}}$$

$$x = X \wedge y = Y \Rightarrow ?P \quad \frac{\begin{array}{l} \{?P\} \\ t := x; \\ (x := y; \\ y := t) \end{array}}{\{x = Y \wedge y = X\}}$$

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$$\begin{array}{l} \{?P\} \\ t := x; \\ \{?R_1\} \end{array}$$

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Larger Programs



$$\begin{array}{l} \{?R_1\} \\ x := y \\ \{?R_2\} \end{array}$$

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$$x = X \wedge y = Y \Rightarrow ?P$$

$$\{x = Y \wedge y = X\}$$

Larger Programs



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Larger Programs



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Larger Programs



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Larger Programs


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Larger Programs



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$$x = X \wedge y = Y \Rightarrow ?P$$

$\{?P\}$

$t := x;$

$\{?R_1\}$

$\{?R_1\}$

$x := y$

$\{?R_2\}$

$\{?R_2\}$

$y := t$

$\{x = Y \wedge y = X\}$

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$$x = X \wedge y = Y \Rightarrow ?P$$

$\{?P\}$

$t := x;$

$\{?R_1\}$

$\{?R_1\}$

$x := y$

$\{?R_2\}$

$\{?R_2\}$

$y := t$

$\{x = Y \wedge y = X\}$

$\{x = X \wedge y = Y\}$

$t := x;$

$x := y;$

$y := t$

$\{x = Y \wedge y = X\}$

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$$x = X \wedge y = Y \Rightarrow ?P$$

{?P}

t := x;

{?R₁}

{?R₁}

x := y

{?R₂}

{?R₂}

y := t

{x = Y ∧ y = X}

{x = X ∧ y = Y}

{?P}

t := x;

{?R₁}

x := y;

{?R₂}

y := t

{x = Y ∧ y = X}

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$$\{x = X \wedge y = Y\}$$
$$t := x;$$
$$x := y;$$
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$$t := x;$$
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$$\{?R_1\}$$
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Larger Programs



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$$\{x = X \wedge y = Y\}$$
$$\{y = Y \wedge x = X\}$$
$$t := x;$$
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$$x := y;$$
$$\{x = Y \wedge t = X\}$$
$$y := t$$
$$\{x = Y \wedge y = X\}$$
$$\{P[E/x]\} x := E \{P\}$$
$$x = X \wedge y = Y \Rightarrow y = Y \wedge x = X ?$$

Yes. QED

Skip Rule



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Statements

$S \quad := \quad \mathbf{skip} \mid v := E \mid a[NE] := E \mid p(E_1, \dots, E_n) \mid S_1; S_2 \mid$
 $\quad \mathbf{if } B \mathbf{ then } S_1 \mathbf{ else } S_2 \mathbf{ endif} \mid \mathbf{while } B \mathbf{ do } C \mathbf{ done}$

where v is a variable, a is an array variable, and p is a procedure name.

$$\{ \quad \} \mathbf{skip} \{ \quad \}$$

Skip Rule



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Statements

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 $\mathbf{if } B \mathbf{ then } S_1 \mathbf{ else } S_2 \mathbf{ endif} \mid \mathbf{while } B \mathbf{ do } C \mathbf{ done}$

where v is a variable, a is an array variable, and p is a procedure name.

$\{ \quad \} \mathbf{skip} \{ \quad \}$

How does the **skip** statement,
which does nothing,
change what is/isn't true?

Skip Rule



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Statements

$S \quad := \quad \mathbf{skip} \mid v := E \mid a[NE] := E \mid p(E_1, \dots, E_n) \mid S_1; S_2 \mid$
 $\mathbf{if } B \mathbf{ then } S_1 \mathbf{ else } S_2 \mathbf{ endif} \mid \mathbf{while } B \mathbf{ do } C \mathbf{ done}$

where v is a variable, a is an array variable, and p is a procedure name.

$$\{ \quad \} \mathbf{skip} \{ P \}$$

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Statements

$S \quad := \quad \mathbf{skip} \mid v := E \mid a[NE] := E \mid p(E_1, \dots, E_n) \mid S_1; S_2 \mid$
 $\mathbf{if } B \mathbf{ then } S_1 \mathbf{ else } S_2 \mathbf{ endif} \mid \mathbf{while } B \mathbf{ do } C \mathbf{ done}$

where v is a variable, a is an array variable, and p is a procedure name.

$$\{ \quad \} \mathbf{skip} \{ P \}$$

How does the **skip** statement,
which does nothing,
change what is/isn't true?

It doesn't

Skip Rule



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Conditional Rule



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$$\{ \quad \quad \quad \} \textbf{ if } B \textbf{ then } S_1 \textbf{ else } S_2 \textbf{ endif } \{ Q \}$$

Conditional Rule



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$$\frac{\{P_1\} S_1 \{Q\},}{\{ \quad \} \text{ if } B \text{ then } S_1 \text{ else } S_2 \text{ endif } \{Q\}}$$

Conditional Rule



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$$\frac{\{P_1\} S_1 \{Q\}, \{P_2\} S_2 \{Q\}}{\{ \quad \} \text{ if } B \text{ then } S_1 \text{ else } S_2 \text{ endif } \{Q\}}$$

Conditional Rule



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$$\frac{\{P_1\} S_1 \{Q\}, \{P_2\} S_2 \{Q\}}{\{(B \Rightarrow P_1) \wedge (\neg B \Rightarrow P_2)\} \textbf{if } B \textbf{ then } S_1 \textbf{ else } S_2 \textbf{ endif } \{Q\}}$$

Example



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$\{ \quad \}$
if $x > y$ **then**

$r := x;$

else

$r := y;$

endif

$\{r \geq x \wedge r \geq y\}$

Example



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{true}

if $x > y$ **then**

$r := x;$

else

$r := y;$

endif

$\{r \geq x \wedge r \geq y\}$

Example



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Apply consequence rule first

{true}

if $x > y$ **then**

$r := x;$

else

$r := y;$

endif

$\{r \geq x \wedge r \geq y\}$

Example



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Apply consequence rule first

$$\frac{P' \Rightarrow P, Q \Rightarrow Q', \{P\} S \{Q\}}{\{P'\} S \{Q'\}}$$

{true}

if $x > y$ **then**

$r := x;$

else

$r := y;$

endif

$\{r \geq x \wedge r \geq y\}$

Example



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{true}

{?P}

if $x > y$ **then**

$r := x;$

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endif

$\{r \geq x \wedge r \geq y\}$

Example



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{true}

{?P}

if $x > y$ **then**

$r := x;$

else

$r := y;$

endif

$\{r \geq x \wedge r \geq y\}$

Example



$\{\text{true}\}$
 $\{?P\}$
if $x > y$ **then**

$r := x;$

else

$r := y;$

endif

$\{r \geq x \wedge r \geq y\}$

$$\frac{\{P_1\} S_1 \{Q\}, \{P_2\} S_2 \{Q\}}{\{(B \Rightarrow P_1) \wedge (\neg B \Rightarrow P_2)\} \text{ if } B \text{ then } S_1 \text{ else } S_2 \text{ endif } \{Q\}}$$

Example



{true}
 $\{(x > y \Rightarrow ?P1) \wedge (x \leq y \Rightarrow ?P2)\}$

if $x > y$ **then**

 {?P1}

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else

 {?P2}

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

Example



{true}

$\{(x > y \Rightarrow ?P1) \wedge (x \leq y \Rightarrow ?P2)\}$

if $x > y$ **then**



{?P1}

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else

{?P2}

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

Example



{true}

$\{(x > y \Rightarrow ?P1) \wedge (x \leq y \Rightarrow ?P2)\}$

if $x > y$ **then**



{?P1}

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else



{?P2}

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

Example



{true}

$\{(x > y \Rightarrow ?P1) \wedge (x \leq y \Rightarrow ?P2)\}$

if $x > y$ **then**



{?P1}

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else



{?P2}

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

$$\frac{}{\{P[E/x]\} x := E \{P\}}$$

Example



{true}

$\{(x > y \Rightarrow x \geq x \wedge x \geq y) \wedge (x \leq y \Rightarrow y \geq x \wedge y \geq y)\}$

if $x > y$ **then**

$\{x \geq x \wedge x \geq y\}$

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else

$\{y \geq x \wedge y \geq y\}$

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

Example



{true}

$\{(x > y \Rightarrow x \geq x \wedge x \geq y) \wedge (x \leq y \Rightarrow y \geq x \wedge y \geq y)\}$

if $x > y$ **then**

$\{x \geq x \wedge x \geq y\}$

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else

$\{y \geq x \wedge y \geq y\}$

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

Example



? {true}
 $\{(x > y \Rightarrow x \geq x \wedge x \geq y) \wedge (x \leq y \Rightarrow y \geq x \wedge y \geq y)\}$

if $x > y$ **then**

$\{x \geq x \wedge x \geq y\}$

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else

$\{y \geq x \wedge y \geq y\}$

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

Example



✓ $\{ \text{true} \}$
 $\{ (x > y \Rightarrow x \geq x \wedge x \geq y) \wedge (x \leq y \Rightarrow y \geq x \wedge y \geq y) \}$

if $x > y$ **then**

$\{x \geq x \wedge x \geq y\}$

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else

$\{y \geq x \wedge y \geq y\}$

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$

Example



{true}

$\{(x > y \Rightarrow x \geq x \wedge x \geq y) \wedge (x \leq y \Rightarrow y \geq x \wedge y \geq y)\}$

if $x > y$ **then**

$\{x \geq x \wedge x \geq y\}$

$r := x;$

$\{r \geq x \wedge r \geq y\}$

else

$\{y \geq x \wedge y \geq y\}$

$r := y;$

$\{r \geq x \wedge r \geq y\}$

endif

$\{r \geq x \wedge r \geq y\}$